15

The future of HACCP

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Although the system that we know as HACCP has been around for 30 years now, there was little development of the original Pillsbury/NASA/US Natick army concept until the late 1980s/early 1990s. In just over a decade since then the HACCP system has undergone a transformation from its original concept to one that is now codified, accepted internationally as the most appropriate means of ensuring food safety, and implemented to various degrees, on a global basis. Just over a decade ago it would have been difficult to predict the changes that lay ahead. Similarly, it may be difficult to predict with much degree of accuracy those changes to HACCP that will occur in the foreseeable future. However, some forces for change are clearly on the horizon now, and it is worth looking at these to try to judge what the future of HACCP is likely to be.

There are a number of forces for change that will impact upon HACCP in the years to come. These include:

- the increasing globalisation and harmonisation of trade between countries
- the changing role of governments and regulatory authorities in the assessment of HACCP
- the role of HACCP in new science/food safety initiatives such as Quantitative Risk Analysis
- the need for application of HACCP throughout the supply chain.

In addition to the specific forces for change there will always be the ongoing requirement for the continuing improvement of the HACCP system based on practical experience. The problems highlighted in the preceding chapters of this book certainly point to this need.

15.1 Globalisation and harmonisation of trade

The introduction to this book touched on some of the implications for HACCP resulting from the globalisation of trade, the World Trade Organisation (WTO), and the Sanitary and Phytosanitary (SPS) agreement. The increasing globalisation of trade has focused government/regulatory attention on the issue of equivalency in food safety and quality management. As international food trade continues to grow (currently in excess of $500 billion per year), it has become obvious that no one single country is completely self-sufficient in terms of its food needs. The projected growth in world population from 6 billion in 1999 to an estimated 8–10 billion in 2050, with the majority of such growth being in developing countries, means that there will be a significantly increased demand for more food.

More efficient, and therefore more industrialised, food production will be required with the potential for more movement in foodborne pathogens and more widespread food safety problems. More people will be immunocompromised and will be more sensitive to foodborne pathogens. There will be an increasingly urgent need to improve control over food safety to prevent widespread, globalisation of foodborne disease arising from globalisation of the food supply. There will be an ever-increasing need to control the safety of both imported and exported foods from all countries, to protect the consumer from unsafe food and to prevent the dumping of unsatisfactory quality food. HACCP has become the universal system of choice for the control of foodborne hazards for the majority of food manufacturing countries, as shown by the increasing adoption of both voluntary and mandatory HACCP systems for both internationally and domestically consumed products in many countries.

The SPS agreement, which focuses on safety issues, requires that measures taken by individual countries are:

- applied only to the extent necessary to protect human health
- based on scientific principles
- not maintained without scientific evidence
- based on an assessment of the risk to health that is appropriate to the circumstance.

Whilst the SPS agreement recognises standards, guidelines and recommendations promulgated by Codex Alimentarius, WTO member countries that adopt Codex standards as sanitary measures do not need to justify these measures. However, the fact that Codex promulgates HACCP as the food safety system of choice means that HACCP will be used as the benchmark process by which food safety is ensured in practical terms. The implications of this are significant. Not only does this require a universally agreed *modus operandi* for HACCP to enable harmonised application, but it also requires an agreed mechanism to judge the acceptability, or not, of HACCP implementation and effectiveness in controlling foodborne hazards. The former requirement is met by the Codex Alimentarius 1997 benchmark HACCP publication (Codex, 1997), but the latter point brings us to the issue of equivalence.

Article 4 of the SPS agreement states:

> Members shall accept the sanitary or phytosanitary measures of other Members as equivalent, even if these measures differ from their own or from those used by other Members trading in the same product, if the exporting Member objectively demonstrates to the importing Member that its measures achieve the importing Members' appropriate level of sanitary or phytosanitary protection. For this purpose, reasonable access shall be given, upon request, to the importing Member for inspection, testing and other relevant procedures.
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> Members shall, upon request, enter into consultations with the aim of achieving bilateral and multilateral agreements on recognition of the equivalence of specified sanitary and phytosanitary measures.

The concept of equivalence basically means that an exporting country need not apply the same control measures for processing as those used in the importing country, provided the degree of food safety assurance delivered is deemed equivalent to that of the importing country. This also applies to the inspection system used by both countries (Orriss, 1999). The concept of equivalence is one of the most contentious issues in food safety at the present time. Whilst some countries have already begun negotiation of equivalency agreements, there is still much debate concerning the principles underlying equivalency and precisely how it is possible, in practical terms, to determine the equivalency of imported products. Codex Alimentarius is currently seeking agreement on a better definition of equivalency. At the present time it is likely that the determination of equivalence will involve consideration of three tiers of information:

- the infrastructure within a country
- the laws and regulations with respect to food safety
- Food Safety Objectives/HACCP/SPS agreement.

The concept of Food Safety Objectives will be discussed in a later section, but it seems most likely that determination of equivalency will have to include, amongst others, a consideration of the HACCP system currently applicable to the food/country involved. HACCP seems certain therefore to play a key role not only as the food safety system of choice but also as a key element in the determination of equivalency of the food safety systems between countries.

Hathaway (1999) provides a good description of the issues surrounding the judgement of equivalence and the use of Food Safety Objectives. A key step for the future will be to agree practical guidelines for the determination of equivalency internationally. Such an agreement would still leave open the question of monitoring compliance (done by the importing country?), and question the role of initiatives such as HACCP certification. Will an internationally recognised standard for HACCP certification have a role to play as a measure of HACCP system effectiveness, or will a more complex system be required to assess the effectiveness of a multifaceted Level of Protection/FSO/

HACCP system? It is also possible to identify where conflicts of interest could arise in the future. The challenge of maintaining appropriate standards of food safety (equivalence) in the face of political pressures to allow/increase freedom of trade will be one of the biggest challenges facing Codex and national regulatory agencies in the future.

15.2 Changing role of governments and regulatory bodies

National governments have a responsibility to ensure consumer health and safety by introducing and implementing legislation relating to food safety. In terms of specific measures such as HACCP, government agencies have a strategic role to play in the implementation of HACCP and a regulatory/enforcement role to play in the assessment of industry-owned HACCP systems. The recent WHO/FAO Consultation on Guidance on Regulatory Assessment of HACCP (WHO/FAO, 1998) addressed the issues of how regulatory agencies should assess HACCP systems. The Consultation recognised that government agencies need to show leadership by promoting and facilitating the implementation of HACCP. The type of activities that government agencies need to consider include:

- facilitating training programmes for industry
- providing guidance, expertise and, where necessary, legislation
- encouraging the development of training material for industry, government and other interested parties
- formulating a programme to assess HACCP.

As the implementation of HACCP systems continues to increase on a world-wide basis, the traditional role of the food safety agencies, including that of food inspectors, is having to change to adapt to the new ways of industry working. In addition to the inspection of industries for compliance with GMP and other relevant guidelines, officials are also now faced with new responsibilities for the assessment of HACCP systems. There is no doubt that government/regulatory agencies need to rethink the way in which they assess food safety control and the above Consultation was intended to provide government agencies with guidelines on regulatory assessment, focusing specifically on:

- the role of government agencies and their responsibilities with regard to HACCP assessment
- the essential activities which need to be carried out when assessing HACCP
- how to perform essential activities adequately.

The implementation of HACCP with its emphasis on product safety and use of CCPs, provides both industry and regulatory bodies with a more logical method of ensuring the safety of foods than was previously available. Whilst industry, as developers and owners of HACCP systems, has developed its own approaches to assess the effectiveness of in-house HACCP systems over a period

of time, government and regulatory agencies now find that they need to consider how they can assess HACCP systems from a regulatory point of view. In this regard government and regulatory agencies should learn from the experience gained within industry on the appropriate methodology to assess HACCP systems (including GMP/prerequisite requirements). Annex 3 of the Codex Consultation report contains a description of an approach that has proved successful in the industry assessment of HACCP.

One of the major problems that government and regulatory agencies have to address is that, because HACCP is a more science-based system of food safety control than those previously used, the majority of government/regulatory assessors do not, themselves, currently have the requisite skills and competencies to assess HACCP systems satisfactorily. There is therefore a need for assessors to receive education in such areas as hazard analysis as well as training in HACCP principles, much of which must be of a practical nature, in order for them to be capable of assessing HACCP systems with confidence. Even this may not be enough. Assessors may need to get involved actively in HACCP studies (experiential learning) in order for them truly to appreciate the approach.

Another major hurdle for government/regulatory agencies to overcome lies in the amount of time and the nature of the resource required to assess HACCP systems satisfactorily. Whilst some HACCP systems in SMEs can be assessed in a short time period if the assessor has the right skills and competencies (see above), many HACCP systems in large manufacturing plants require significant amounts of time from the assessor for a thorough evaluation. This requirement is currently in conflict with the amount of skilled resource available to government/regulatory agencies. This is potentially a serious problem.

Assuming the assessors have the appropriate skills and competencies, a further key task is to facilitate an assessment process that is searching and meaningful yet can be completed within a relatively short time frame. An answer lies in the role of validation. The recent ILSI Monograph on the Role of Validation and Verification (ILSI, 1999), clarifies the different roles of validation and verification well. Validation is, basically, concerned with ensuring that the scientific and technical content of the HACCP plan is correct, and should be carried out before HACCP plan implementation. Verification is about ensuring that the HACCP plan owners do what they say they will do in the HACCP plan and this should be carried out on an ongoing basis. In addition, at regular but infrequent intervals (e.g. one year), a full reappraisal of the HACCP plan should occur.

A summary of evidence gathered during validation would therefore provide a government/regulatory assessor with data upon which the scientific and technical competency of the HACCP plan could then be judged. A summary of the evidence gathered during verification would similarly provide the assessor with data upon which could be judged the rigour with which the HACCP plan was implemented. Summaries of validation and verification would therefore seem to offer assessors data upon which, with the requisite skills and competencies, they would be able to make a judgement on the plan's scientific

and technical content, and the rigour of its implementation. Such an assessment could be made within one day even for a complex HACCP plan. For this initiative to become a reality, both industry and government/regulatory assessors have a responsibility to work together and agree a process that satisfies the requirements of both groups.

Validation and verification should be applied in SMEs as well as large manufacturing plants. In SMEs the evidence for validation and verification can and should be as simple as possible, otherwise it will not be collected. Whilst the principles of validation and verification should be applied sympathetically to SMEs, there are genuine problems for SMEs in terms of the availability of time and skilled resource to carry out hazard analysis. Government and regulatory bodies have an important role to play in the mentoring of SMEs to help fill the gaps in technical knowledge and other resources. In this regard the use of industry sector specific 'generic' HACCP plans can, with guidance, prove very valuable in getting SMEs started and later as a check that what they have achieved is in line with industry expectations.

Similarly, the subsidised training of SMEs in HACCP plan development and implementation can contribute significantly to overcoming the barrier to start HACCP that SMEs find so hard to overcome themselves. This help, together with ongoing support, is one of the real opportunities that government/regulatory agencies have in making a significant improvement in food safety standards on a worldwide basis. However, the fact that government/regulatory agencies can have a dual role of both mentoring HACCP implementation and assessing HACCP system effectiveness means that there is a potential conflict of interest. It is therefore essential that assessors do not assess HACCP systems that they also helped to design and/or implement.

15.3 The role of HACCP in new science/food safety initiatives

It is now clear that HACCP is only part of the series of events and processes required for food safety, regulatory compliance and customer satisfaction. One of the problems that HACCP suffers from is that it is really a system of food safety hazard identification and control and the application of HACCP rarely, if ever, considers the reduction in risk to the consumer. A typical criticism of HACCP is that it is not outcome oriented and it is therefore difficult to assess its benefit. As Orris and Whitehead (2000) point out, a hazard is defined as a biological, chemical of physical agent in, or condition of food with the potential to cause an adverse health effect. However, the term 'potential' is subjective and is interpreted broadly from remotely possible to highly probable. Orris and Whitehead also point out that there is often confusion about what constitutes a hazard, leading to further inconsistency in HACCP application.

The challenge is to assess the application of HACCP in the context of its contribution towards eliminating or reducing the risk of foodborne illness to the consumer. The practicalities underpinning control of food safety leading to

measurable reduction in consumer risk is currently a major goal for both industry and governments alike and is invoking considerable scientific debate. Space does not permit a detailed review of all current thinking, but the following discussion will illustrate the role that HACCP may play in the future in this important development.

The SPS agreement requires members to ensure that their SPS measures are based on a risk assessment, taking into account techniques developed by relevant international organisations. The Codex Alimentarius Commission (CAC) has prepared a draft document on Risk Analysis guidelines relevant to the above. Risk Analysis as defined by Codex consists of the following three processes:

1. risk assessment
2. risk communication
3. risk management.

Risk assessment is the science-based process that provides data for use in risk management decision making. The application of risk analysis to food microbiological hazards is a significant challenge due to the lack of good quality quantitative data on topics such as exposure assessment, but increasingly quantitative risk analyses (QRA) are being published for major pathogens/food types. As knowledge increases we should expect more QRA studies to be published. It is proposed that the outcome of a risk assessment should be a Level of Protection (LOP) (e.g. estimated number of cases per year in a certain food per 100,000 of the consumer population). Risk management should then be used to decide if this LOP is acceptable or appropriate (ALOP). The outcome of QRA should therefore be an acceptable (appropriate) level of protection (ALOP). The ALOP is a level of consumer food safety risk that would then be adopted as food safety policy by the national government.

For international trade there is an implicit obligation of WTO members to determine their ALOP. The current thinking is that Food Safety Objectives (FSOs) will then need to be developed to represent the ALOP. Food Safety Objectives are intended to translate a level of risk (ALOP) to a level of hazard. A working definition of the term FSO would be: 'a statement based on a risk analysis process which expresses the level of a hazard in food that is tolerable in relation to an ALOP.'

An example of an FSO would be: '$<$ 100 L. monocytogenes per gramme of food at the moment of consumption.' This example of an FSO is both measurable and verifiable and could be used to judge equivalence. The role of HACCP would be as the practical food safety control system, implemented on the factory or kitchen floor, required to deliver food product complying with the above FSO. HACCP would still stay as an industry owned, practical system of hazard control, but it would now be outcome oriented and would have to meet the FSO. As such, HACCP systems would then be linked to consumer risk via the FSOs and the relevant ALOP. The concept of FSOs would also allow for the assessment of the stringency of different HACCP systems. There is no doubt that

currently HACCP systems covering the same hazard/product combinations in different plants/companies/countries differ in their stringency because they have been developed without the benefit of having to meet a measurable outcome. The use of FSOs would allow the comparison of different HACCP systems addressing the same hazard/product combination.

It is important to reiterate that the above discussion is currently the subject of much scientific debate and there are conflicting views from those summarised here. It is also true to say that many of the practical issues have yet to be resolved. However, whilst the specific details may change with the passage of time and scientific debate, the authors of this chapter are convinced that the future role of HACCP will be as part of a total food safety system incorporating many, if not all, of the concepts outlined above. Such a move would strengthen the role of HACCP significantly by making it outcome oriented, but would still leave ownership of HACCP as the responsibility of industry as it is now. Most importantly, it maintains the practical 'hands on' nature of HACCP as it is now, and does not impose an unnecessary additional burden of scientific assessment on the development of HACCP plans.

Many SMEs already find it difficult to carry out the current requirements for hazard analysis on their own and require additional technical support. It is crucially important for both present and future users of HACCP that we do not impose additional burdens (e.g. risk assessment) on HACCP planning, otherwise a substantial part of the very good work done over the last 30 years will be wasted. The above scenario maintains the correct balance between HACCP practicality and science and has much to commend it.

15.4 Application of HACCP further along the supply chain

The history of HACCP shows clearly that the major thrust for HACCP development came initially from the large manufacturers and subsequently from regulatory interests. The current CAC benchmark document clearly reflects the skills, competencies, technical knowledge, etc. of the larger manufacturing organisations. As Barnes has pointed out in Chapter 12, the concept of teams of experts and detailed product-by-product examination may be relevant for manufacturing plants, but it is not well suited to smaller establishments (SMEs) such as caterers handling large numbers of products. Taylor and Route have also pointed out in their chapters many of the problems faced by SMEs in trying to implement HACCP. A recent WHO (1999) consultation examined HACCP implementation studies for small and less developed food businesses, and there is hope that this and other related initiatives will lead to a more appropriate HACCP approach for SMEs.

What the early history of HACCP illustrates, however, is that HACCP development has been quite narrowly focused. There is a need to extend the application of HACCP, or HACCP principles, beyond the food manufacturing industry to cover the whole food chain from 'farm to fork' more effectively.

There is an increasing awareness that the whole of the food chain needs to have safety as a priority. In this regard the codification of HACCP, and its perceived ownership by large manufacturers and regulators, probably does not help. However, the authors of this chapter believe that HACCP is applicable to all sectors of the food chain.

The principles that underlie HACCP (i.e. the identification of significant hazards and associated critical limits and control measures to ensure control of relevant hazards) can be applied to any sector of the food chain, provided that the principles are applied in different ways depending upon the business type. The practical application of HACCP principles should be tailored to the particular food sector/business size under consideration, enabling the philosophy underlying HACCP to be applied without all the 'baggage' that comes with the full codified version that is most applicable to large manufacturing organisations.

Barnes makes the point in his chapter that this flexibility of approach needs to be more formally recognised by Codex. Provided that a flexible approach is used there is no reason why HACCP cannot be applied successfully on the farm, in slaughterhouses, in primary crop production, in retail and food service sectors as well as manufacturing. Some businesses are doing this but it needs to become commonplace. There remain important issues to be resolved (e.g. the lack of a killing step CCP in slaughter operations), but the future of HACCP should embrace the widespread application of its principles across the entire food supply chain.

In a similar vein, it could be argued that as HACCP implementation spreads across the globe it is most likely that differences in local culture, native industries, and locally accepted standards could lead to very fragmented application of the currently accepted HACCP system. One could ask if this is likely to be a problem if it happens. The answer is given above, that provided the philosophy underpinning HACCP is the same, the practical application of the principles should allow flexibility. Codex Alimentarius has a key role to play in this regard, and should be seeking specifically to endorse a flexibility of approach that will enable a genuinely inclusive application of HACCP basic principles. This would ensure that global users of HACCP will be confident that HACCP can be made appropriate to their needs, and in so doing, prevent the development of 'HACCP alternatives' or 'different versions of HACCP'.

15.5 The continued development of HACCP

The above discussion has allowed us to speculate how four specific forces for change could affect the future of HACCP. In addition to these specific issues there will always be other, more general issues which will require HACCP to evolve in order to maintain its effectiveness. There is an increasing realisation that HACCP is not a static system that can be 'put on the shelf' once the HACCP

plan has been finalised, but a system continually evolving in the face of new challenges and in the light of practical experience.

15.5.1 New/emerging hazards

One of the disturbing features of food safety has been the identification of a number of new and/or emerging hazards that have occurred over the last decade. Microorganisms such as *Eschericia coli* 0157:H7, *Listeria monocytogenes*, and *Salmonella enteritidis* have all led to foodborne death or illness. Similarly there is heightened awareness and concern in the USA and Europe about the increase in allergic reactions to food. HACCP has had to evolve to accommodate such hazards. This ability of HACCP to evolve as a result of new challenges and improve as a result of practical experience is an essential feature behind the success of HACCP to date, and it will continue to be essential for the future success of HACCP. One interesting consideration for the future is that currently food safety control is focused on acute microbiological hazards. Recent research has indicated that some foodborne microorganisms may be associated with chronic disease. If this subsequently is proven to be the case then HACCP, as the food safety system of choice, will be required to adapt to include consideration of hazards with chronic effects. This will involve specific consideration of chronic effects during the hazard analysis process.

HACCP may also be extended to encompass hazards not previously associated with HACCP systems. As an example, Kane makes a very interesting point in Chapter 4 that, from the retailers' point of view, food manufacturers should consider malicious product contamination as a food safety hazard. Kane makes the point that HACCP principles can be specifically focused on this issue and control measures implemented to reduce the opportunity for deliberate product contamination, just as retailers have had to deal with the prevention of malicious contamination in their stores. In reality it is believed that very few manufacturers include malicious contamination as a realistic hazard in their HACCP plans, rightly or wrongly because they believe the risk to be low. It is of course possible when considering own label brands that the perception of such risk is viewed to be much lower by the manufacturer than by the retailer who, should such a contamination incident occur, would have to face the immediate consequences.

Kane also makes a very interesting point concerning post-launch monitoring (PLM). Kane's point is that data from customer complaint monitoring that retailers carry out could include specific post-launch monitoring of new products, looking more closely for any emerging food safety problems. Such data could be shared with product manufacturers and fed back into future product launches and HACCP systems. Use of PLM data in this way would presumably fall under the current HACCP terminology of verification. As such, PLM, whether carried out by the retailer or the manufacturer, has the potential to improve the quality of both current and future HACCP plans significantly.

It is not uncommon to hear people at international conferences say 'this is an aspect of HACCP that I was not aware of'. Such statements do not demonstrate a

lack of consistency or a weakness in the HACCP system. On the contrary they demonstrate the essential ability of HACCP to continue to evolve in response to new food safety challenges. HACCP practitioners should never believe that once the HACCP plan is prepared then the job is finished. HACCP is a long-term commitment and practitioners must review and verify their HACCP plans on a regular basis to ensure they remain appropriate.

15.5.2 Food safety management systems

The future of HACCP must also take into account the roles played by several other food safety management systems. Management systems such as Total Quality Management (TQM), the ISO 9000 series, Good Manufacturing Practice (GMP), Good Hygiene Practice (GHP), Prerequisites, Quality plans, etc. will all have some impact on the future of HACCP. In effect, HACCP must form a key element of an integrated approach to food management. HACCP, underpinned by GMP/Prerequisites, should form those elements of the programme targeted at food safety (ILSI, 1998). The food safety management programme itself is a component of the company's larger Quality Management System (e.g. ISO 9000 series), and long-term managerial strategy (TQM). In this way it is possible to see clearly the role that HACCP will play as part of a total integrated management plan. It is clear that HACCP must not be viewed as a stand-alone system for, without the proper underpinning support of GMP/Prerequisites, HACCP itself cannot hope to function effectively, and without integration with other food management tools, the specific role of HACCP mediated product safety cannot be defined clearly within a company's overall management strategy.

15.6 Summary

When we talk about the future of HACCP it is important to remember that even in larger companies, which were the first to implement HACCP, the systems are still relatively new, and in some medium-sized and smaller-sized companies HACCP principles are still being implemented for the first time. It is also important to remember that the speed of HACCP implementation is not uniform across the globe. Industries in many countries are currently tackling HACCP for the first time. There is plenty of scope for HACCP implementation in some countries/regions/sectors of the food business to continue to develop up to the point already reached in the USA/Europe/Canada/Australia/New Zealand, etc. For some, the future of HACCP can be seen clearly in the progress already made by others, and this in itself would be a tremendously valuable step forward for many. In those countries where HACCP is well accepted there is still much to do, particularly in extending the system further down the supply chain to truly cover 'farm to fork' and in further collaboration between industry and regulatory agencies to ensure valid, cost-effective assessment of industry-owned HACCP systems.

It seems certain that HACCP will play an increasingly important role as the food safety system of choice for food manufacturers and regulatory agencies and also as the system of choice when considering the safety of foods traded internationally. HACCP will continue to evolve in response to new food safety challenges and as a result of experience in use. A particularly exciting development will involve the use of HACCP in conjunction with new concepts of acceptable level of protection and food safety objectives. Such a development offers the promise that HACCP will become outcome oriented and will be clearly linked to a reduction in consumer foodborne disease. Such developments must be the subject of open, transparent debate, preferably under the guidance of Codex Alimentarius, and must include representatives from all sectors and sizes of industry as well as regulatory agencies. All developments in the HACCP system must maintain the essential 'hands on' practical nature of HACCP, remembering that HACCP principles must be applicable in the factory, slaughterhouse, butchers shop, etc. and must be applicable to large, medium-sized and small industries. The imposition of unnecessary science on the HACCP system must be avoided at all costs.

Not all future developments in HACCP will be adopted at a uniform pace globally, nor need they be. Codex has an important role to play in ensuring that, whilst the principles of HACCP remain the same for all, there is flexibility in the practical application of such principles. Countries/companies/industries that are now experienced in the implementation and assessment of HACCP have a responsibility to pass on to others the lessons learnt, both good and bad. Safe food is the aim of all; there should be no competitive advantage in having a 'better HACCP plan'. It is essential that we all learn from the past. We hope the contents of this book will go some way towards ensuring that the essential lessons from HACCP implementation and assessment are available to a wider audience.

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